

*Article type*

## Development of an improvising machine in an ubimus context

---

### Abstract

This paper explores the relationship between design decisions in improvising machines and the ideological foundations of ubimus movement practices. To this end, we present a *free/libre* open-source Pure Data interactive system, named “XXXXXXXXXX”, initially designed to evaluate the potential of learning free improvisation through interaction with improvising machines in the context of social distancing experienced [in 2020](#). Our system is now envisioned as a possible creative tool with applicability in areas other than improvisation, such as composing electronic music, extending digital instruments, and its use in aesthetic proposals that are not necessarily related to free improvisation and will be discussed in further detail. [We believe our](#) system's free software and do-it-yourself approach [could](#) inspire musicians to explore the creative potential of decentralized technological interaction, [which](#) is especially relevant [when](#) intelligent agents, often poorly understood and controlled, are increasingly pervasive in our lives.

**Keywords:** ubimus; interactive music systems; improvising machines; free improvisation; computer creativity.

---

### 1. [Introduction](#)

The ubiquitous music movement (ubimus) arises from the convergence of musical creativity, ubiquitous computing tools, and the social dynamics of participant interaction, redefining how music is created and experienced<sup>[1]</sup>. Ubimus has been developing from its initial characteristics by integrating new musical activities and perspectives that may even challenge the necessity of technological support. As a result, the definitions of ubimus have been updated regularly. Recent publications suggest that a consensus for a definition has been solidified in the “notion of ubimus as an ecology or ecosystem of stakeholders and resources that support creative endeavors”<sup>[2]</sup>.

While not circumscribed as a ubimus practice, previous initiatives in the area have been related to the genre of free improvisation. Stolfi *et al.*<sup>[3]</sup> describes the development of an online music-

making tool that takes advantage of the Freesound library<sup>1</sup> to make real-time selections and combinations of Creative Commons audio samples in the context of live improvisations. In another example, Clemente *et al.*<sup>[4]</sup> conducted a series of experiments with Information and Communication Technologies (ICT) and Ubiquitous Computing (UC) as tools to support free improvisation practices of a saxophone ensemble, observing improvements in the group's sound quality, according to the students and teachers who took part in the experiment.

Free improvisation can be understood as a musical practice associated with an anti-formalist approach to music-making, where musical pieces are improvised at the moment, and practitioners often abstain from the explicit use of conventional musical structures and techniques in favor of the exploration of sonic materials that arise from the interplay between musicians. The difficulty of studying free improvisation, evidenced by the obstacles found in suggestions for its formalization<sup>[5]</sup>, suggests that other paths are necessary for its understanding and, ultimately, apprehension. While free improvisation avoids predetermined structures, performers still require preparation for public performances, and as we understand it, many improvisers allude to the practice of repeated ensemble performances as the most fruitful methodology for musicians to delve into the free improvisation universe<sup>[6-9]</sup>.

This scenario begs the question: if ensemble practice proves to be the primary learning approach for free improvisation, how can musicians without access to companionship for improvisation sessions achieve some level of proficiency in this genre? In this sense, the pandemic world of 2020 has offered a challenging scenario for musicians who want to practice in-person musical activities. However, social distancing also opened doors to the exploration of connected forms of musical creation for a broader pool of participants, not previously engaged in this sort of musical experience. Technological solutions proposed by companies for the *musical internet* typically try to encompass a wide range of musical applications for a single product, often reinforcing the acoustic-instrumental model of musical production and a hierarchical approach to musical interaction. In the other direction, Messina et al.<sup>[10]</sup> stress the ability of ubimus ecosystems to adapt their actions to local conditions, leveraging the flexibility of available resources and approaching the presence of technological resources not just as passive instruments but as active stakeholders in the creative process.

---

<sup>1</sup> Available at <https://freesound.org/>.

Tatar and Pasquier<sup>[11]</sup> define *musical agents* as systems that “explore the notions of autonomy, reactivity, proactivity, adaptability, coordination and emergence” in interaction with an environment of musical creativity, ultimately reducing their creative tasks into twelve categories. This paper aims to draw connections between the reasoning behind the development of *improvising machines*, one of those categories, and the ideological principles underpinning ubimus practices, while pointing to the use of this kind of musical agent as a plausible companionship to learn free improvisation. To do so, we discuss the latest decisions in the design, implementation, and future perspectives for our improvising machine, named [REDACTED], a free/libre open-source software<sup>2</sup> (FLOSS) project developed in Pure Data (Pd), a graphical programming environment designed by Miller Puckette for real-time audio processing<sup>3</sup>, widely employed by computer music enthusiasts since its launch in the nineties.

This work aims to contribute to future projects that seek to integrate the use of artificial agents into the ubimus movement, through a theoretical review of perspectives from free improvisation and ubimus, and a technical presentation of the system implemented. We believe that the do-it-yourself (DIY) approach and the open distribution nature of our system are capable of helping other musicians in the endeavor to create new creative solutions in both fields.

## 2. The system

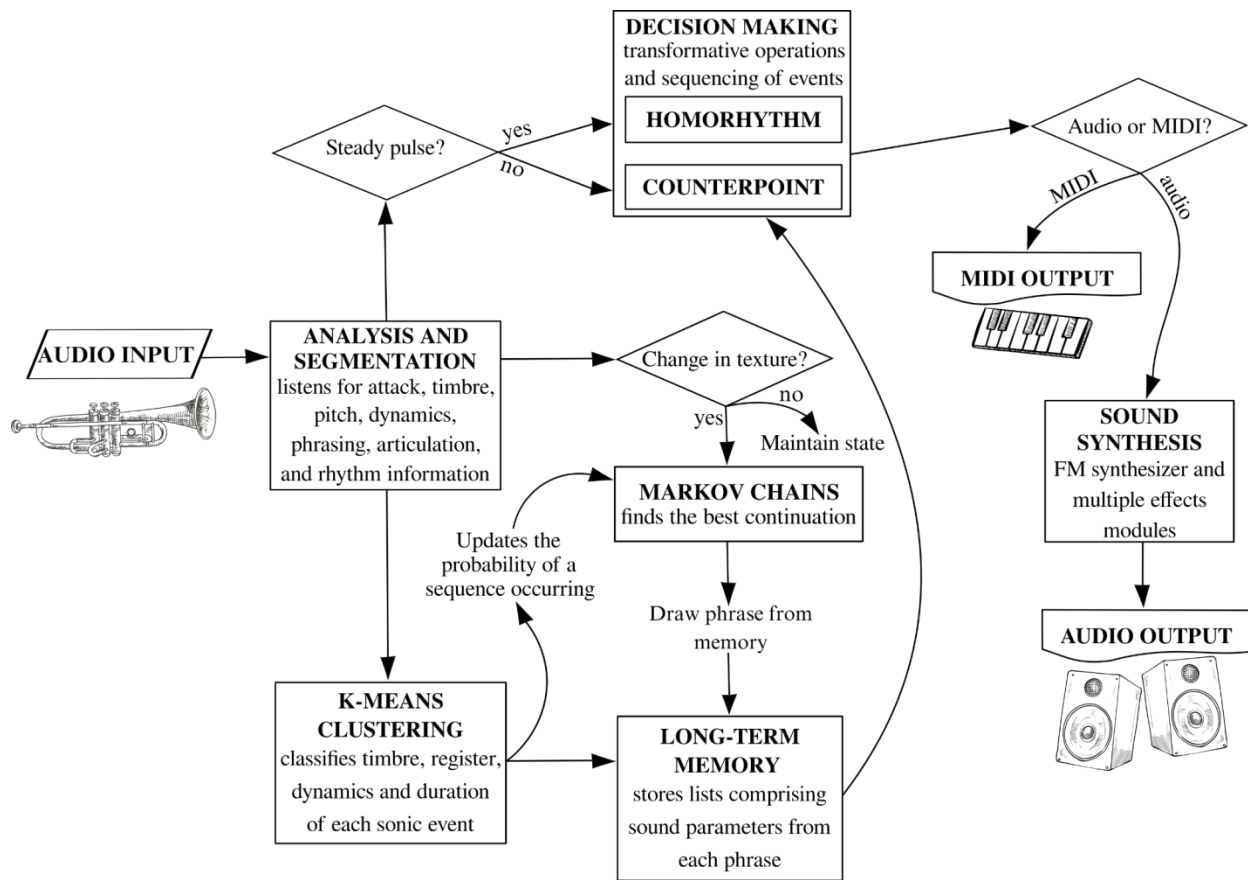
[REDACTED] is a musical agent originally designed for live performances of free improvisation and approached in previous work<sup>[12]</sup> as a feasible companionship for practicing the genre. This section presents its later version, with novel solutions for each step of the interaction.

Like other musical agents for live performances, the interaction model begins with [REDACTED] analyzing the audio input of the musician. Each note or sonic event played by the musician is broken into many musical parameters, and the sequence of events is segmented into “phrases” to be classified and stored in memory. To interact with the musician, the system selects a phrase that corresponds with each parameter, manipulates its values with mathematical operations and recombines all parameters into sonic events for the construction of new musical discourses. The

<sup>2</sup> Code available at [https://github.com/\[REDACTED\]](https://github.com/[REDACTED]).

<sup>3</sup> Download Pd and access its documentation at <https://msp.ucsd.edu/software.html>.

sonic responses of the machine can be delivered as real-time synthesized audio or MIDI messages to be sent to any digital music instrument. A simplified diagram of this architecture is shown in Figure 1.



**Figure 1.** Diagram depicting the architecture of the implemented system.

Before starting an improvisation, the musician must press a button to measure the background noise intensity in the room. Sounds played by the musician which surpass this threshold will be interpreted as “events”, while moments of sound intensity that fall below this threshold are interpreted as “pauses”. In the context of free improvisation, a musical genre that is not based on the use of functional harmony and can employ nonstandard melodic and rhythmic phrasing, it would be difficult to assess the end of a musical phrase from traditional methods of analysis. To overcome this difficulty, it was needed to establish another parameter for identifying musical phrases. Therefore, in **XXXXXXXX**, the average pause duration between the last four events is used as a threshold to identify the beginning and end of a phrase. Thus if, after a series of sound events, the musician pauses for a longer time than the threshold, for example, to take a breath,

the system interprets this longer pause as a moment of “silence”, grouping the previous sound events into a single phrase. Each musical parameter extracted from a sound event is added to separate lists. Therefore, a musical phrase is disassembled into several lists.

We believe that a decision-making system for an improvising machine must not only react to what is being played at the moment but also be able to predict where the improvisation is heading. To facilitate this process, each parameter of a sound event is classified among possible predefined clusters, and the predominant cluster found in the phrase is used to classify the phrase as a whole, regarding that parameter. This process is performed by the *k-means* algorithm, an unsupervised clustering method widely used in data science, which aims to categorize  $n$  observed points into a number  $k$  of non-overlapping groups predefined by the user. The algorithm searches for the centroid of each group  $k$ , assigning each data point to the group whose centroid is closest. The process can be repeated several times (according to the number of iterations defined by the user) and the algorithm leans to clusters of similar size. In XXXXXXXXXX, datasets for “training” were provided for each sound parameter, and the algorithm is used to categorize each new of new observation into predefined clusters.

The musical parameters timbre, register, dynamics, and duration are categorized with *k-means* into varying numbers of clusters which we believe could potentially represent a simplified perception of one dimension of these musical parameters by musicians. Each parameter of a sonic event receives a numerical value to represent their cluster, corresponding to: (0) non-percussive and (1) percussive sound for timbre; (0) low, (1) middle, and (2) high registers; (0) piano, (1) mezzo and (2) forte dynamics; and (0) long, (1) moderate and (2) short durations. Lists comprising the cluster identified for each sound event are recorded in distinct datasets<sup>4</sup>, which constitute collections of all lists of parameters played by the musician during past and current improvisation sessions, essentially functioning as long-term memory. To match a phrase to a specific dataset, we use the statistical mode found for the phrase, following the same clusters for sound events<sup>5</sup>, approximating how a musician might perceive prevalent aspects of a musical texture. For example, in a sequence of many short notes and one long note, the prevailing characteristic will be understood as short durations.

---

<sup>4</sup> Simple .csv files.

<sup>5</sup> Thus providing two datasets for timbre, and three for pitch, intensity and tempo each.

As soon as the identified cluster for a musical parameter changes, meaning that the musician has altered some prevalent aspect of the musical texture being played, [REDACTED] randomly selects a list from a dataset corresponding to the same parameter. To decide which dataset to choose from, the system makes use of *second order* Markov chains to model the likelihood of the cluster for the following phrase. As suggested by Roads<sup>[13]</sup>, Markov chains are a class of statistical sequencing algorithms that calculate the probability of occurrence of the next value in the sequence considering only the state in which the system is located (zeroth order). For musical applications, considering the last state (first order), or even the last two states (second order), would enable the construction of sequences with a more organized structural sense. As the order increases, the material generated by the algorithm approaches the original content of the sequence. This method has been periodically used in musical agents that perform imitative tasks, statistically modeling the playing style of a particular human whom it interacts with, generally concerning rhythmic and melodic aspects<sup>[14]</sup>.

In turn, [REDACTED]'s use of Markov chains does not serve the purpose of “imitating” what is played by the user phrase level, but rather looking for trends in behavior between each phrase played. For example, once a musician repeats a pattern, such as playing notes in the high register, followed by notes in the low register, the machine will tend to replicate this behavior with greater probability, looking for a phrase in the respective dataset (in the example case, a dataset of phrases in the low register). However, the complete musical texture may differ greatly from the material played by the musician, since each musical parameter of the original phrase has been disassociated from the others. Also, as will be discussed later, each list of parameters used to generate the new phrase is individually modified by transformative mathematical operations, a process that aims to generate variations on the original material, alluding to methods of composition and improvisation commonly used in different musical genres to generate new musical material from a single source.

Initially programmed with five independent voices, [REDACTED] now only employs a monophonic texture to respond to the user. The choice to simplify the interaction to just one voice aims to direct the user's listening experience to more concise musical content, making it easier to understand what the machine is playing. Two processes for transforming the musical material

drawn from memory can be used to generate new content of the musical response: *homorhythm* and *counterpoint*.

A homorhythmic texture<sup>6</sup> is activated when the system identifies a low standard deviation for the last few inter-onset intervals (IOI) played by the musician<sup>7</sup>. Even though sophisticated algorithms have been proposed in the literature for beat identification, this simple method has been effective in quickly perceiving whether the musician is approaching the literal reproduction of a pulse, allowing the machine to promptly join the beat. Although handled by a different algorithm, Collins<sup>[15]</sup> has proposed, similarly, a rhythmic analysis to “differentiate free time playing from steadier metrically established performance”, inspiring the decision for the two modes of texture interaction implemented in [REDACTED]. Musical parameters for each note played during the homorhythmic texture are extracted not from memory but from the last phrase played, bringing the machine's sound material closer to what has just been played by the human.

When no steady pulse is identified, [REDACTED] plays a contrapuntal texture, in which lists of parameters drawn from memory are manipulated and combined to formulate new phrases to be played by the machine. As each sound parameter is dissociated from the others found in its original phrase, and transformative operations are applied to the list, the resulting texture is unlikely to be heard as directly related to a phrase played by the musician. To manipulate each list of parameters, the contrapuntal method randomly selects from a range of mathematical functions to be applied to the numerical values of the list taken from memory. The functions for rhythmic manipulations and their explanations can be seen in Table 1. The system represents notes by their pitch classes<sup>[16]</sup>(PC), and the functions used to manipulate them are presented in Table 2. Finally, Table 3 presents the functions for amplitude manipulations.

After each list of parameters is transformed by these functions, the contrapuntal method combines the parameters into a new list, defining the parameters for the next note, to be sent to the synthesizer. Other parameters, not taken from memory but directly from the last phrase

<sup>6</sup> That is, when the machine makes sounds that start at the same time as the sounds made by the musician with whom it interacts.

<sup>7</sup> The value used as reference was fine tuned by trial and error.

played, include a legato index<sup>8</sup> and a spectral flatness curve<sup>9</sup>, directly replicating elements from the last phrase played by the musician. The use of these additional parameters brings both the articulation and the sound roughness closer to what is played by the musician in the short term.

**Table 1.** Functions that can be applied to rhythmic manipulation of stored phrases.

|                    |                                                                               |
|--------------------|-------------------------------------------------------------------------------|
| <b>Same</b>        | Bypasses the original list with no modification.                              |
| <b>Accelerando</b> | Rearranges the list of durations in descending order of values.               |
| <b>Ritardando</b>  | Rearranges the list of durations in ascending order of values.                |
| <b>Shuffle</b>     | Shuffles the order of the durations in the original list.                     |
| <b>Multiple</b>    | Multiplies the duration values by a random factor.                            |
| <b>Addition</b>    | <i>Added Value Rhythm</i> as described by Messiaen <sup>[18]</sup> .          |
| <b>Ostinato</b>    | Chooses a duration and repeats it for the number of items in the list.        |
| <b>Rotation</b>    | Rotates the list for a random number of iterations.                           |
| <b>Reversion</b>   | Reverses the list's order.                                                    |
| <b>Random</b>      | Creates a new list of random values, with boundaries taken from the original. |
| <b>Markov</b>      | Chooses values from the list with a second order Markov chain.                |

**Table 2.** Functions that can be applied to pitch manipulation of stored phrases.

|                   |                                                                         |
|-------------------|-------------------------------------------------------------------------|
| <b>Same</b>       | Bypasses the original list with no modification.                        |
| <b>Ascending</b>  | Rearranges the list of PC in ascending order of values.                 |
| <b>Descending</b> | Rearranges the list of PC in descending order of values.                |
| <b>Transpose</b>  | Transpose pitches in the original list by a random number of semitones. |
| <b>Multiply</b>   | Multiplies the PC values by a random factor.                            |

<sup>8</sup> *Legato* refers to a playing style where notes are smoothly connected without breaks. In this context, the term “legato index” refers to a metric to quantify the smoothness or connectivity between sound events heard by the machine. This index is calculated from the proportion of silence between two notes, not identified as a pause.

<sup>9</sup> *Spectral flatness* measures the noisiness of a spectrum by comparing its geometric and arithmetic mean. Low values suggest a tonal spectrum, such as a sinusoid, while high values indicate a flat, noisy spectrum<sup>[17]</sup>.



|                              |                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------|
| <b>Complement</b>            | Replaces the original PC set with its complement <sup>10</sup> .                  |
| <b>Ascending Complement</b>  | Rearranges the list of the complementary set of PC in ascending order of values.  |
| <b>Descending Complement</b> | Rearranges the list of the complementary set of PC in descending order of values. |
| <b>Transpose Complement</b>  | Transpose PC in the complementary set by a random number of semitones.            |
| <b>Multiply Complement</b>   | Multiplies the PC values of the complementary set by a random factor.             |
| <b>Markov</b>                | Chooses values from the PC set with a second order Markov chain.                  |

**Table 3.** Functions that can be applied to amplitude manipulation of [stored](#) phrases.

|                    |                                                                        |
|--------------------|------------------------------------------------------------------------|
| <b>Same</b>        | Bypasses the original list with no modification.                       |
| <b>Crescendo</b>   | Rearranges the list of intensities in ascending order of values.       |
| <b>Decrescendo</b> | Rearranges the list of intensities in descending order of values.      |
| <b>Equal</b>       | Chooses a duration and repeats it for the number of items in the list. |

A Frequency Modulation (FM) synthesizer receives the list containing the parameters set for the next sound event (timbre, pitch, [dynamics](#), duration, [articulation](#), and [spectral flatness curve](#)), combining these defined aspects with some degree of randomness in the synthesis process to [enable](#) subtle variations between each sound [heard](#) in a phrase. The output of the FM synthesizer can be passed through a randomly selected sound effect, further modifying the resulting sound. The effects [currently implemented](#) are ring [modulation](#), pitch [shift](#), delay lines, [granulation](#), low-pass and high-pass filters, amplitude [modulation](#), and [reverberation](#). The parameter settings for each sound effect use [values from](#) the same list of parameters received by the [synthesizer](#), [combined with](#) randomized values for each event.

<sup>10</sup> As defined by Forte<sup>[16]</sup>, when receiving a pitch class set equivalent to the C major scale (the white notes of the piano), for example, its complementary set would be the F sharp major pentatonic scale (the black notes of the piano), which represents all the pitch classes that remain for the universal set of twelve notes to be reached.

Network improvisation sessions can also be held using [REDACTED]. At the moment, this functionality makes it possible to interact aurally with a second user and their version of the system, geographically distant, in real-time, with very low [latency](#), by using *Quacktrip*<sup>11</sup>, a Pure Data implementation of the *JackTrip* audio transmission protocol<sup>12</sup>. Although ubimus initiatives are not necessarily dependent on proposals based on synchronous musical interaction<sup>[10]</sup>, we note that Rottondi *et al.*<sup>[20]</sup> suggest that situations of instrumental interaction can tolerate delays of between twenty and fifty milliseconds. According to Bosi *et al.*<sup>[19]</sup>, *JackTrip* allows latencies below two hundred milliseconds for connections made between points more than ten thousand kilometers apart, allowing for viable synchronicity between two parties if this is in their interests.

### 3. Ubimus and improvising machines

In this section, we explore the similarities between musical perspectives in the ubimus movement and those found in the principles behind the creation of musical agents, especially those designed for [the context of](#) free improvisation. Examples of characteristics found in [REDACTED] and other improvising machines will illustrate these relationships.

First, the ubimus movement aims to integrate musical and sonic materials, challenging the traditional separation found in many classical and electronic music practices<sup>[2]</sup>. Similarly, many improvising machines, especially those developed for the context of free improvisation, seek to bring together timbral and symbolic elements as aspects of equal relevance to sound creation and interaction with the system. For example, systems developed by Hsu<sup>[21]</sup> benefit from an analysis of spectral roughness measured from sound input alongside intensity and tempo to identify different “modes” of playing. Another relevant approach by Collins<sup>[15]</sup> employs k-means clustering to identify ten different sonic scenarios based on the timbre content of the last few seconds. Nort *et al.*<sup>[22]</sup> favored recognizing gestural and textural content in their free improvisation machine, emphasizing sonic elements as the musical basis for the interaction.

Furthermore, Lewis<sup>[23]</sup> highlights that improvising machines should be viewed as active participants in music-making, rather than passive tools controlled by composers. This

<sup>11</sup> To refer to the documentation, see <http://msp.ucsd.edu/tools/quacktrip/doc.htm>.

<sup>12</sup> [Jacktrip is an network communication protocol by Chris Chafe that enables low-latency uncompressed audio interconnections for musicians who wish to play together remotely.](#)

perspective aligns with ubimus principles, emphasizing non-hierarchical collaboration. A similar point of view is supported by the ubimus movement when authors indicate that “a paradigm of musical interaction targeting passive ‘instruments’, ‘orchestras’ or enforcing centralized decision-making may ignore key aspects of this ubimus ecosystem”<sup>[24]</sup> and that in “the context of ubiquitous systems, a device is not a passive object that a musician can play. It is an agent in a dynamical system that adapts itself to the musical activity, to the local environment, and to the other agents that interact with it”<sup>[1]</sup>. Illustrating this idea, the orchestra developed by Lewis in his early renditions of *Voyager* allowed for the independence of decisions made by each of the 64 voices available in the system<sup>[25]</sup>, eliminating the centralizing role of a conductor in favor of a model of multifaceted collective musical production, such as that found in the encounters of the AACM collective of which he was a fundamental participant<sup>[23]</sup>.

Many improvising machines, as well as [REDACTED], seek to analyze and manipulate symbolic elements of the material played by a musician with whom it interacts to produce their behavior. This paradigm, for Lewis, represents an interaction technique that is “essentially Eurocentric in nature”, as reintroductions of material that has already been played “come into conflict” with the direction of improvisation<sup>[26]</sup>. Contrasting to this method, old renditions of *Voyager* employed the analysis of the material played by the musician only as a “filter” for the system’s decisions, restricting itself to listening to a short-term memory consisting of only the last five notes played by the musician.

This issue is addressed in [REDACTED] by disassociating each parameter from the others found in a sonic event, reassembling them in novel contexts that differ from the original phrase, and also transforming the material through mathematical operations. The decision-making and synthesis processes are also nondeterministic, applying second-order Markov chains to navigate the memory of recorded phrases and randomness to create small sonic variations. The reappearance of any musical material, therefore, cannot be controlled by those who play with the system, who at most act as an influence on the results.

Keller *et. al*<sup>[27]</sup> identify ubimus practices under the methodological framework of sonic ecologies: “habitats where agents and objects interact producing creative sonic by-products”, while Keller and Lazzarini<sup>[28]</sup> stress that ubimus depend not only on musical and sound resources but on the relationships established between peers and with the surrounding environment. Free

improvisation, while a practice particularly focused on the act of collective performance, embraces adaptive behaviors of participants to the audience and particular sonic characteristics of a venue. Improvising machines developed for this context can be thought of as agents capable of establishing creative relationships with the sonic environment that differ from those assembled by human performers.

Ubumus also embraces approaches to music-making that do not depend on acoustic instrumentation models, traditionally associated with corporeal movement and sometimes seen as the “only valid strategies for musical interaction”<sup>[1]</sup>. This paradigm, in Keller’s<sup>[29]</sup> view, ends up imposing interaction modes and processes developed and maintained by the European classical music tradition. Even though a large population within the genre of free improvisation still relies on the use of traditional acoustic instruments<sup>[9]</sup>, their interaction approaches and instrumental techniques often question those traditionally associated with classical music, allowing not only the presence of unusual techniques and sound sources but permitting the interaction of music with other forms of artistic expression. For example, Keith Rowe, the guitarist for the group AMM, reports abandoning his guitar technique by ceasing to tune it and preferring to manipulate it laying it down on a table with objects of various kinds. The process was inspired by how Jackson Pollock abandoned traditional painting techniques in favor of the “drip technique”, as Rowe himself comes from a visual arts background<sup>[29]</sup>.

When dealing with technological interfaces, Keller<sup>[29]</sup> also notes that the design of ubimus tools should provide intuitive interfaces, avoiding “cognitive overload” for users who are already busy maintaining their creative proposals. Observing this perspective, the development of  endeavored to make it easier for users to adjust basic settings needed to start a performance: its interface makes it possible to prepare and begin an improvisation session with just three mouse interactions. Similarly, an online session can be initiated with just two more mouse clicks.

Finally, as discussed by Scott Thomson<sup>[31]</sup>, free improvisation questions the role of the improviser as a virtuoso, contradicting a historical perspective of European concert music very recurrent in jazz cultures. In the same way, ubimus is based on creative approaches that embrace “little-c musical phenomena that were excluded from professional musical practices”<sup>[28]</sup>, questioning the preeminence of virtuosity in music-making.

#### 4. Discussion and future perspectives

This section addresses previous concepts and prospects for [REDACTED], based on the theoretical relationships between improvising machines and the ubimus movement, presented in the previous section.

The system was originally designed as an artificial companionship for practicing free improvisation in individual contexts. The idea comes from a practical concern brought up by the experience of social distancing in 2020, when the likelihood of gathering a full ensemble to improvise remained remote for months. Even though solo performance is not unheard of in the free improvisation scene<sup>[6]</sup>, some authors question whether the practice shouldn't be understood as real-time composition<sup>[9]</sup>, as the interaction between peers seems to be as important, if not even more significant, than the musical material being played<sup>[8]</sup>. Alternatives to ensemble practice have been presented in a few textbooks. Tom Hall<sup>[33]</sup>, for example, proposes both dealing with musical parameters separately, and encompassing all aspects of free playing in exercises that can be done individually, in pairs, or in groups. Crook<sup>[34]</sup>, although in a book oriented towards free jazz, proposes that it is necessary to learn the basics of the jazz idiom to truly understand what free playing is.

The interest in instructional applications for musical agents is not new. As Rowe<sup>[34]</sup> and Lewis<sup>[25]</sup> explain, designing with such systems might provide us with a meaningful understanding of how we approach music interaction. In this sense, the original proposition for [REDACTED] was to provide musicians who did not have human companionship to practice free improvisation with a computer system capable of interacting within this context. In previous informal evaluation scenarios, experienced improvisers who interacted with the system have judged it to be aesthetically effective for the genre of free improvisation. To continue the research in this direction, plans are underway to implement further evaluation processes with a more formal approach.

Besides the investigation of the instructional value of the implemented machine, we believe [REDACTED] also demonstrates the potential for diverse applications in creative endeavors which are suggested next. First, we address the possible inclusion of audience participation and/or environmental resources, where distinct sonic and relational ecosystems from those of free improvisation might be achieved. For example, future developments could incorporate sonic

relationships with the environment into the system's audio analysis, such as processing and replaying sounds below the “event identification” thresholds. This approach, overlooked in the current design, aligns with ubimus perspectives by challenging the binary division between “desired” and “undesired” sounds in traditional music<sup>[32]</sup>. Another perspective could be brought by allowing that, if the system is employed in a “concert” scenario, which it was not originally designed to, other modalities of interaction could be implemented by Local Area Network (LAN) connectivity, like admitting the participation of an audience to exchange sonic or symbolic data with the machine, influencing in the aesthetic results of the interaction.

In addition, Lewis<sup>[36]</sup> points out that the distinction between the activity of “composition” and “improvisation”, for a computer program, does not exist. So how do we perceive the difference between the two perspectives? The notion of algorithmic composition, for him, arises from the idea that a composition program's output is not arbitrary but limited to producing only the “correct” results. In this sense, the use of [REDACTED] for other initiatives focused on a compositional approach could also be explored. Since a recording feature is already available for both sounds played by the musician, as well as the sounds produced by the machine, each improvisation with the system is capable of generating sonic material to be exploited in electroacoustic composition proposals. The unexpected results of the interaction with a machine might also yield musical ideas to be explored in compositions of instrumental or interactive pieces.

Although the preferred model for interaction in free improvisation is usually by directly approaching sound relations, possibilities can also be explored through the MIDI output provided by [REDACTED]. For example, instruments that allow both the reproduction of sounds through their speakers and the exchange of MIDI data with a computer, such as most commercially available digital keyboards, could be explored in pieces whose system response uses the instrument played by the musician. Modifications to [REDACTED]'s response algorithms could facilitate the perception of this proposal as an “extended instrument” in which the control is not entirely in the hands of the player, with varying degrees of autonomy.

The same proposals can be imagined for other FLOSS musical agents that permit their users the art of modifying their code, allowing for creative outcomes unforeseen by their original developers. The DIY nature of our project can be viewed as both an inspiration and an invitation

to interested stakeholders to take part in the development of novel creative applications. As the presence of commercial solutions that employ the use of artificial intelligence (AI) becomes increasingly ubiquitous, there is a need to reflect on the impact of the closed-source and centralized nature that some for-profit companies offer them. Creative proposals of a FLOSS nature offer a different road for addressing the interaction with intelligent systems, functioning not only as a "one size fits all" tool but also allowing its adaptation to different social contexts and their specific local resources.

## 5. Conclusion

This paper explored the intersection of the ubimus movement and improvising machines, presenting [REDACTED], an interactive system designed for free improvisation. By examining the theoretical relationships between these domains, we illustrated how both seek to break down traditional boundaries in music-making, embracing non-hierarchical, adaptive, and collaborative approaches to creativity.

Both ubimus and free improvisation rely on dynamic relationships between agents, peers, and the surrounding environment, creating ecosystems where interaction and adaptation are key. Hence, we envision that improvising machines could have the potential for various artistic endeavors in both scenarios. [REDACTED], designed for accessibility and adaptability, already facilitates innovative approaches to improvisation and other creative musical activities involving artificial agents. By taking advantage of network connectivity, we have a broader scope of possible interactions between this project and other ecosystems in the *musical internet*.

In conclusion, [REDACTED] offers a model for creating adaptive, non-hierarchical, and community-oriented musical environments. Its FLOSS nature enhances its potential for widespread use and continual improvement, encouraging users to innovate and adapt the system according to their unique desires and experiences, in contrast with the closed and often opaque commercial AI solutions that have been dominating the creative technology landscape.

## References

1. Flores, L. V., Pimenta, M. S., Capasso, A., Tinajero, P., and Keller, D. (2009). Ubiquitous music: concepts and metaphors. *Simpósio Brasileiro de Computação Musical*, 158-169.
2. Bridges, B., Keller, D., Messina, M., and Yaseen, A. (2023). Ubiquitous Music as a Movement? (UbiMus 2023 Editorial). *UbiMus2023 Proceedings*, V-X.
3. Stolfi, A., Milo, A., Viola, F., Ceriani, M., and Barthet, M. (2018). Playsound. space: An ubiquitous system in progress. *VIII UbiMus Workshop Proceedings*, 54-65.
4. Clemente, M., Falleiros, M., Tavares, T., and Fornari, J. (2020). Experiments on technology-assisted free improvisational practices with an ensemble of saxophone. *X Ubiquitous Music Workshop*, 121-132. <https://doi.org/10.5281/zenodo.4247685>
5. Linson, A. (2014). Investigating the cognitive foundations of collaborative musical free improvisation: Experimental case studies using a novel application of the subsumption architecture [PhD Thesis, Open University (United Kingdom)]. Open Research Online.
6. Bailey D. (1993). *Improvisation: Its Nature and Practice in Music*. Da Capo Press.
7. Borgo, D. (2005). *Sync or swarm: Improvising music in a complex age*. A&C Black.
8. Stenström, H. (2009) *Free ensemble improvisation* [PhD Thesis, Academy of Music e Drama; Höskolan för scen och musik].
9. Corbett, J. (2016) *A Listener's Guide to Free Improvisation*. University of Chicago Press.
10. Messina, M., Freitas, B., Simurra, I., Gómez, C., Aliel, L., Keller, D. (2023). Musical stuff, technologically convergent and disruptive factors. *Proceedings of the International Symposium on Ubiquitous Music, 2023*, 74-85. 13th International Symposium on Ubiquitous Music 2023 (UbiMus23), Ulster University, Derry~Londonderry, Northern Ireland.
11. Tatar, K. and Pasquier, P. (2019). Musical agents: A typology and state of the art towards musical metacreation. *Journal of New Music Research*, 48(1):56–105. <https://doi.org/10.1080/09298215.2018.1511736>



- 
13. Roads, C. (1996). *The Computer Music Tutorial*. MIT Press.
14. Pachet, F. (2003). The continuator: Musical interaction with style. *Journal of New Music Research*, 32(3), 333-341.
15. Collins, N. (2011). Ll: Listening and learning in an interactive improvisation system. *Unpublished Report*. University of Sussex.
16. Forte, A. (1973). *The Structure of Atonal Music*. Yale University Press.
17. [Lerch, A. \(2022\). An introduction to audio content analysis: Music Information Retrieval tasks and applications. John Wiley & Sons.](#)
18. Messiaen, O. (1944). *The Technique of my Musical Language*. Alphonse Leduc.
19. Bosi, M., Servetti, A., Chafe, C., and Rottondi, C. E. M. (2021). Experiencing remote classical music performance over long distance: A jacktrip concert between two continents during the pandemic. *Journal of the Audio Engineering Society*, 69(12):934–945.
20. Rottondi, C. et al. (2016). An overview on networked music performance technologies. *IEEE Access*, IEEE, 4:8823–8843. <https://doi.org/10.1109/ACCESS.2016.2628440>
21. Hsu, W. (2008). Two approaches for interaction management in timbre-aware improvisation systems. *International Conference on Mathematics and Computing*.
22. Nort, D. V., Oliveiros, P., and Braasch, J. (2013). Electro/acoustic improvisation and deeply listening machines. *Journal of New Music Research*, 42(4):303–324.
23. Lewis, G. E. (1999). Interacting with latter-day musical automata. *Contemporary Music Review*, 18(3):99–112.
24. Keller, D., Costalonga, L., and Messina, M. (2020). Ubiquitous music making in covid-19 times. *Proceedings of the 10th Workshop on Ubiquitous Music (UbiMus 2020)*.
25. Lewis, G. E. (2000). Too many notes: Computers, complexity and culture in "voyager". *Leonardo Music Journal*, 10:33–39.

26. Dean, R. (2003). Hyperimprovisation: computer-interactive sound improvisation, 19. AR Editions, Inc.
27. Keller, D., Lazzarini, V., and Pimenta, M. S. (2014). Ubimus through the lens of creativity theories. *Ubiquitous music*, 3–23. Springer.
28. Keller, D., and Lazzarini, V. (2017). Ecologically grounded creative practices in ubiquitous music. *Organised Sound*, 22(1), 61–72.
29. Keller, D. (2020). Everyday musical creativity. *Ubiquitous Music Ecologies*, 23–51. Routledge.
30. Warburton, D. (2001). Keith Rowe interview. Available from: <https://www.paristransatlantic.com/magazine/interviews/rowe.html>. [Last accessed: 2024 Nov 16]
31. Thomson, S. (2007). The pedagogical imperative of musical improvisation. *Critical Studies in Improvisation/ Études critiques en improvisation*, 3(2).
32. Messina, M., and Aliel, L. (2019). Ubiquitous music, gelassenheit and the metaphysics of presence: Hijacking the live score piece ntrallazzu 4. *Proc. of the 14th International Symposium on CMMR*, 685-695.
33. [Hall, T. \(2009\). Free improvisation: A practical guide. Bee Boy Press.](#)
34. [Crook, H. \(2006\). Beyond Time and Changes: A Musician's Guide to Free Jazz Improvisation. Advance Music.](#)
35. [Rowe, R. \(2004\). Machine musicianship. MIT press.](#)
36. Taylor, G. (2020). An Interview with George Lewis and Damon Holzborn, Part 1. Available from: <https://cycling74.com/articles/an-interview-with-george-lewis-and-damon-holzborn-part-1>. [Last accessed: 2024 Nov 17]